

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - MEDIUM (Assignment)

COURSE CODE: CSA0904

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 3

Total Marks: 30

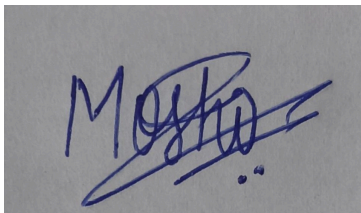
ANNEXURE A

Assignment

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Moshika', with a stylized flourish extending from the end.

ANNEXURE

Assignment

1. ATM PIN Security Validator

"A nationalized bank receives ATM PINs from customers through a kiosk. Develop a Java program that accepts the PIN as a string and performs the following operations:

Remove unwanted spaces using trim()

Validate that the PIN contains exactly 4 digits using matches()

Display the PIN length using length()"

2. Employee ID Normalization System

"An HR department receives employee IDs in mixed formats such as emp1023. Create a Java program to:

Remove spaces using trim()

Convert the alphabetic prefix to uppercase using toUpperCase()

Join the prefix and number using concat()"

3. Online Shopping Coupon Verification

"An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()

Converts the entered code to uppercase using toUpperCase()

Compares it with the original coupon using equalsIgnoreCase()"

4. Student Email Username Extractor

"A college portal stores student emails such as arun23@college.edu.

Develop a program that:

Finds the position of @ using indexOf()

Extracts the username using substring()

Displays the username length using length()"

5. Restaurant Digital Order Analyzer

"A restaurant stores orders as DOSA-3-IDLY-2-VADA-1. Create a program that:

Splits the order using split()

Converts food names to uppercase using toUpperCase()

Counts the number of ordered items using length()"

6. Railway Coach and Seat Parser

"Indian Railways stores ticket information as S5-43. Write a Java program that:

Splits the value using split()

Extracts the coach number using substring()

Displays the coach type using charAt()"

7. Inventory Product Search System

"A supermarket stores product names in different letter cases. Develop a program that:

Converts product names to lowercase using toLowerCase()

Searches a product using contains()

Confirms exact matching using equalsIgnoreCase()"

8. Grocery Shopping Cart Management System

"Add products using add()

Remove purchased products using remove()

Check whether a product exists using contains() Check how many products there in the list"

9. Bank Token Queue Management System

"A bank serves customers based on token order. Using Queue:

Add customer tokens using offer()

Serve the next customer using poll()

Display the upcoming customer using peek()"

10. Railway Parcel Dispatch Tracking System

"A railway parcel office loads parcels onto a dispatch container. The last loaded parcel must be unloaded first, and parcel details must be searchable.

Requirements:

Store parcel IDs in a Stack

Store parcel ID and destination in a HashMap

Unload parcels using pop() and display the destination of the unloaded parcel"

11. Hospital Appointment Scheduling System

"A hospital issues appointment tokens to patients and stores patient details separately.

Requirements:

Add appointment tokens to a Queue

Store token number and patient name in a HashMap

Serve patients in token order and display the served patient details"

12. University Course Enrollment Manager

"A university maintains a list of students enrolled in a course. Students who withdraw must be removed from the list.

Requirements:

Add student names to an ArrayList

Traverse the list using an Iterator

Remove withdrawn students and display the updated enrollment list"

13. Warehouse Damaged Item Removal

"A warehouse stores item names in an inventory list. Damaged items must be removed safely while traversing the collection.

Requirements:

Store inventory items in an ArrayList

Traverse the list using an Iterator

Remove items marked as Damaged and display the remaining inventory"

14. Bank Token Management System

"Customers receive tokens in the order they arrive at a bank counter.

Requirements:

Add customer tokens to a Queue

Serve customers in FIFO order

Display the next customer to be served and the remaining queue"

15. Web Browser Back Navigation

"A web browser stores recently visited pages so that users can navigate backward.

Requirements:

Add visited page URLs using push()

Perform the Back operation using pop()

Display the current page using peek()"

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

CO2-AT2 Assignment

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QUESTIONS

Q1 
Grocery Shopping Cart
Management System
10 marks

Q2 

Q1 **Grocery Shopping Cart Management System**

10 marks

Add products using add()
Remove purchased products using remove()
Check whether a product exists using contains() Check how many products there in the list

Online Shopping Coupon
Verification
10 marks

Q3 
ATM PIN Security Validator
10 marks

3/3 answered

Your Code

```
1 import java.util.ArrayList;
2 public class GroceryCart {
3     public static void main(String[] args) {
4         ArrayList<String> cart = new ArrayList<>();
5         cart.add("Milk");
6         cart.add("Bread");
7         cart.add("Eggs");
8         cart.add("Butter");
9         System.out.println("Cart after adding products: ");
10        System.out.println(cart);
11        String checkItem = "Milk";
12        if (cart.contains(checkItem)) {
13            System.out.println(checkItem + " is present in the cart.");
14        } else {
15            System.out.println(checkItem + " is not present in the cart.");
16        }
17        String checkItem2 = "Cheese";
18        if (cart.contains(checkItem2)) {
19            System.out.println(checkItem2 + " is present in the cart.");
20        } else {
21            System.out.println(checkItem2 + " is not present in the cart.");
22        }
23        System.out.println("Total number of products in the cart: " + cart.size());
24    }
25 }
```

Input

stdin input

Output

```
Cart after adding products:
[Milk, Bread, Eggs, Butter]
Milk is present in the cart.
Cheese is not present in the cart.
Total number of products in the
cart: 4
```


QUESTIONS

Q1Grocery Shopping Cart
Management System

10 marks

**Q2**Online Shopping Coupon
Verification

10 marks

**Q3**

ATM PIN Security Validator

10 marks



3/3 answered

Q2 Online Shopping Coupon Verification

10 marks

An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()

Converts the entered code to uppercase using toUpperCase()

Compares it with the original coupon using equalsIgnoreCase()

Your Code

```
1 import java.util.*;
2 public class CouponVerification {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String coupon = "SAVE50";
6         String code = sc.nextLine();
7         code = code.trim();
8         code = code.toUpperCase();
9         if (code.equalsIgnoreCase(coupon))
10             System.out.println("Coupon Applied Successfully!");
11         else
12             System.out.println("Invalid Coupon Code!");
13     }
14 }
```

Input

SAVE50

Output

Coupon Applied Successfully!



Run



Save



QUESTIONS

Q1

Grocery Shopping Cart
Management System

10 marks

Q2

Online Shopping Coupon
Verification

10 marks

Q3

ATM PIN Security Validator

10 marks

3/3 answered

Q3 ATM PIN Security Validator

10 marks

A nationalized bank receives ATM PINs from customers through a kiosk. Develop a Java program that accepts the PIN as a string and performs the following operations:

Remove unwanted spaces using trim()

Validate that the PIN contains exactly 4 digits using matches()

Display the PIN length using length()

Your Code

```
1 import java.util.*;
2 public class ATMPinValidator {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String pin = sc.nextLine();
6         pin = pin.trim();
7         System.out.println("PIN Length: " + pin.length());
8         if (pin.matches("\\d{4}"))
9             System.out.println("Valid PIN");
10        else
11            System.out.println("Invalid PIN");
12    }
13 }
```

Input

7412

Output

PIN Length: 4
Valid PIN

